

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 2: Inside the Goat Kidney: Cortex, Medulla, Pelvis and Nephrons

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	B	A	B	D	A

USE AFTER: PREDICT PHASE — Predict: sketching the inside of a goat's kidney and Think-Pair-Share prediction

Q1. A learner predicts that a goat's kidney is just a hollow bag that stores urine. Which action would best test this prediction?

- A. Weigh the kidney and compare it with the weight of a full bladder
- B. Cut the kidney lengthwise and observe the inside**
- C. Look only at the colour of the outside of the kidney
- D. Measure the length of the kidney from top to bottom

Correct answer: B. Opening the kidney lengthwise lets learners see whether it is hollow or organised into regions. This is the evidence that confirms or challenges the 'bag of urine' idea.

USE AFTER: OBSERVE PHASE — Observe: dissecting the goat kidney and identifying cortex, medulla and pelvis

Q2. A goat kidney is cut lengthwise. Which region is the darker outer part with a fine, grainy texture?

- A. Ureter
- B. Cortex**
- C. Renal pelvis
- D. Medulla

Correct answer: B. The cortex is the darker outer region and looks grainy because it contains many tiny filtering units (renal corpuscles). The medulla is paler and striped, and the pelvis is hollow.

USE AFTER: OBSERVE PHASE — Observe: examining the external kidney (renal artery, renal vein and ureter at the hilum)

Q3. At the hilum of a kidney, which pair correctly names the vessel that brings blood to the kidney and the tube that carries urine away from it?

- A. Renal vein brings blood; ureter carries urine
- B. Renal artery brings blood; ureter carries urine**
- C. Ureter brings blood; renal artery carries urine
- D. Renal artery brings blood; renal vein carries urine

Correct answer: B. The renal artery delivers blood to the kidney for filtering. The ureter carries urine away, and the renal vein carries blood away from the kidney.

USE AFTER: EXPLAIN PHASE — Explain: correcting the misconception 'the pelvis does not make urine; it collects it'

Q4. In a cut goat kidney, the funnel-shaped hollow space leads into the ureter. What is the main job of this region, the renal pelvis?

- A. It collects urine and funnels it into the ureter**
- B. It stores urine for many hours until the goat urinates
- C. It makes urine by filtering blood
- D. It brings blood into the kidney

Correct answer: A. The pelvis does not make urine. It collects urine from the medulla and funnels it into the ureter.

USE AFTER: EXPLAIN PHASE — Explain: introducing the nephron chart (Bowman's capsule, glomerulus, tubule, loop of Henle, collecting duct)

Q5. Tiny granular dots can be seen in the cortex of a kidney with a hand lens. What is the nephron?

- A. The fatty layer that covers the outside of the kidney
- B. The microscopic functional unit of the kidney**
- C. The funnel-shaped space where urine collects
- D. The tube that carries urine from the kidney to the bladder

Correct answer: B. Nephrons are microscopic filtering units packed in the cortex and medulla, and each one is a functional unit of the kidney. The ureter is the tube that carries urine, and the pelvis is the funnel-shaped space.

USE AFTER: EXPLAIN PHASE — Explain: connecting kidney structure to the goat's concentrated urine and the freshwater fish's dilute urine

Q6. A goat and a freshwater fish live in the same hot area. Which statement best links kidney structure to what each animal does with water?

- A. The goat makes concentrated urine because its pelvis makes urine, while the fish has no kidney
- B. The goat's kidney has no medulla, so it makes dilute urine, while the fish's kidney makes concentrated urine
- C. Both animals make the same urine because all kidneys have the same structure
- D. The goat's kidney has a well-developed medulla that helps it make concentrated urine, while the fish's kidney mainly gets rid of excess water as dilute urine**

Correct answer: D. A well-developed medulla lets the goat's kidney produce concentrated urine. The freshwater fish's kidney mainly removes excess water, so its urine is dilute.

USE AFTER: MODEL BUILDING PHASE — Model: building and labelling the kidney model and checking where blood enters, where filtering happens and where urine leaves

Q7. Which order correctly traces the path of fluid through the regions of a kidney, from where it is filtered to where it leaves?

- A. Cortex, medulla, pelvis, ureter**
- B. Medulla, cortex, ureter, pelvis
- C. Pelvis, medulla, cortex, ureter
- D. Cortex, pelvis, medulla, ureter

Correct answer: A. Blood is filtered in the cortex. The fluid then passes through the medulla into the pelvis and leaves through the ureter. A model of the kidney should show this order.